

2038 — Electrical and Electronics Practice Lab

Semester II

Credits: 1.5

Lab Course

Contact Hours: 45

Official Source Declaration

This lesson is architected based on the official **SITTTR Kerala Revision 2026** syllabus for course code **2038 (Electrical and Electronics Practice Lab)**.

Authoritative Source: SITTTR Syllabus PDF (2038_EE_S2.pdf)

Scope: Covers all 4 Modules, 13 Experiments, and Open-ended experiments as prescribed.

Course Dashboard

Teaching Scheme

- **Practical:** 3 Hours/Week
- **Self-Learning:** 1.5 Hours/Week
- **Total Semester Hours:** 45

Assessment Scheme

- **CIE (Internal):** 60 Marks
- **ESE (External):** 40 Marks
- **Total Marks:** 100

This course provides a hands-on foundation in electrical and electronic systems, covering safety, wiring, protective devices, electronic components, soldering, and Arduino programming.

How to Study This Lesson

- 1 Safety First:** Before touching any equipment, read the safety guidelines in Module 2.
- 2 Theory Sync:** Correlate these experiments with Course 1031 (Basics of Electrical & Electronics Engineering).
- 3 Record Maintenance:** Tabulate all observations accurately as per the CIE evaluation criteria.

Course Outcomes (CO)

CO Code	Outcome Description	Bloom's Level
CO1	Practice basic Electrical Measurements and Wiring Tasks using standard instruments	Apply

CO Code	Outcome Description	Bloom's Level
CO2	Illustrate Electrical Safety Practices with the help of Protection Devices	Apply
CO3	Practice Assembling, connection, testing, and verification of simple electronics circuits on a breadboard and PCB	Apply
CO4	Demonstrate teamwork and problem solving skills in controlling simple electronic circuits with the help of Arduino	Apply

Syllabus Coverage Map

Module	Focus Area	Experiments	Hours
Module I	Measurements & Home Installation	Multimeter, Extension Box, Fan, Inverter	9
Module II	Domestic Wiring & Protection	Distribution Box, Control Panels	9
Module III	Electronics Assembly	Soldering, Rectifiers, Clippers	9
Module IV	Arduino Programming	LED Blinking, Voice Control, Open Ended	18

Learning Roadmap

The course follows a logical progression from **Measurement** to **Installation**, then to **Electronic Hardware**, and finally **Software-controlled Hardware (Arduino)**.

Measurements

Wiring & Safety

Electronics/PCB

Arduino Control

Module I

Electrical Measurements & Home Installations

Instructional Hours: 9 | CO Mapped: CO1

Expt 1: Testing and Measuring Equipment

This experiment focuses on using the **Digital Multimeter (DMM)** and **Clamp Meter**.

- **Multimeter:** Measuring AC/DC Voltage, Resistance, and Continuity.
- **Non-Contact Voltage (NCV):** Detecting live wires without physical contact.
- **Clamp Meter:** Measuring high AC current by clamping around a single conductor (works on the principle of Mutual Induction).

Malayalam support:

ഓൾട്ടിമീറ്റർ ഉപയോഗിച്ച് AC/DC വോൾട്ടേജ്, റെസിസ്റ്റൻസ്, കണ്ടിന്യൂറ്റി എന്നിവ അളക്കുന്നു. നോ-കണ്ടാക്ട് വോൾട്ടേജ് (NCV) ഉപയോഗിച്ച് ശാരീരിക സമ്പർക്കം ഇല്ലാതെ ലൈവ് വയറുകൾ തിരിച്ചറിയുന്നു. ക്ലാമ്പ് മീറ്റർ ഉപയോഗിച്ച് ഹൈ അസിംഗിൾ കറന്റ് അളക്കുന്നു (മുതൽ പ്രതിരോധം).

Expt 2: Extension Box Wiring

Wiring two sockets controlled by separate switches. Key focus: **Wire sizing** and **Earthing**.

Safety Check: Always connect the Phase (Line) wire to the switch. The Neutral wire goes directly to the socket. The Earth wire (Green) must be connected to the larger top pin of the socket.

Expt 3 & 4: Fan Connections & Inverters

Ceiling Fan: Wiring an induction motor fan with a regulator. Modern upgrade: Replacing with a **BLDC (Brushless DC) fan** which is more energy-efficient.

Home Inverter: Connecting a battery (Tall Tubular) to an inverter to provide backup for critical loads like lights and fans.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module II

Domestic Wiring & Protection

Instructional Hours: 9 | CO Mapped: CO2

Expt 5: Distribution Box (DB) Wiring

Familiarization with the path of electricity: **Service Wire → Energy Meter → Main Switch/Isolator → RCCB → MCB → Sub-circuits.**

Device	Full Form	Function
MCB	Miniature Circuit Breaker	Protection against Overload & Short Circuit
RCCB	Residual Current Circuit Breaker	Protection against Earth Leakage & Electric Shock

Expt 6: Control Panels & Isolation

Demonstration of control panels to isolate high power circuits from low power control circuits for safety.

Malayalam support:

രക്ഷാ ഉപകരണങ്ങൾ (RCCB) ഭൂമിയിലേക്ക് വൈദ്യുതി നഷ്ടപ്പെടുന്നത് തടയുന്നു. ഇത് വൈദ്യുതി തൊടുന്നതിൽനിന്നും മനുഷ്യരെ സംരക്ഷിക്കുന്നു. MCB വൈദ്യുതി കമ്പിളിയാകുന്നത് തടയുന്നതിനായി ഉപയോഗിക്കുന്നു.

Module III

Electronics Assembly & Circuits

Instructional Hours: 9 | CO Mapped: CO3

Expt 7: Breadboard & Soldering

- Breadboard:** Used for temporary circuit prototyping without soldering.
- PCB Soldering:** Permanent assembly. Practice de-soldering using a pump or wick.



PCB Tracks Representation

Expt 8 & 9: Rectifiers & Clippers

Rectifiers: Converting AC to DC.

- Half Wave: Uses 1 diode.
- Full Wave (Bridge): Uses 4 diodes.

Clippers: Circuits used to "clip" or remove portions of a waveform above or below a certain level.

Module IV

Arduino Controller & Programming

Instructional Hours: 18 | CO Mapped: CO4

Expt 10-13: Arduino Basics

Introduction to **Arduino UNO** and the **Arduino IDE**.

- **LED Blinking:** Writing the first 'Hello World' of hardware.
- **Alternating LEDs:** Logic for flashing two LEDs.
- **Voice Control:** Interfacing Arduino with a voice module or smartphone to control lights.

Malayalam support:

Arduino പാർക്ക് പരമ്പരയിൽ ഉൾപ്പെട്ട പാർക്ക്. പാർക്ക് പരമ്പരയിൽ ഉൾപ്പെട്ട പാർക്ക്, പാർക്ക് പരമ്പരയിൽ ഉൾപ്പെട്ട പാർക്ക്. IDE പാർക്ക് പരമ്പരയിൽ ഉൾപ്പെട്ട പാർക്ക് പരമ്പരയിൽ ഉൾപ്പെട്ട പാർക്ക്.

Open Ended Experiment

Students must design one experiment independently. Suggested ideas from syllabus:

- 1. Hanging bed switch for lamp control.
- 2. Lamp test holder with fuse.
- 3. 0 to 5V variable DC supply.
- 4. Clap light using Arduino.

Student Activities (Self-Learning)

Week	Activity Description	Hours
1	Chart of IEC vs IEEE electrical symbols	2
2	Market survey: Cost of wires, switches, sockets	1.5
5	Identify components in your home Distribution Box	2
10	Resistor color code examples (3, 4, 5, 6 band)	2
12-14	DIY Arduino Programs (Student's choice)	4.5

Formula & Procedure Bank

Ohm's Law (Context: Resistor selection for LED)

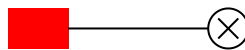
$$R = (V_{\text{supply}} - V_{\text{led}}) / I_{\text{led}}$$

Example: To run a 2V LED from 5V at 20mA: $R = (5-2)/0.02 = 150\Omega$

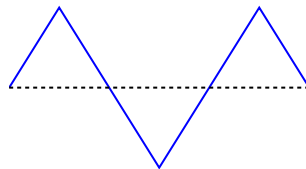
Standard Wiring Colors (India)

- **Phase:** Red / Yellow / Blue
- **Neutral:** Black
- **Earth:** Green or Green-Yellow

Visual Library



Simple Switch-Lamp Circuit



AC Sine Waveform

Assessment Methodology

Component	Method	Marks
CIE (60)	Attendance (5), Continuous Eval (30), CA1 (15), CA2 (15), OEE (10)	60
ESE (40)	End Semester Practical Exam	40

Note: Continuous evaluation includes preparation, setup, observation, analysis, and viva.

Module-wise Q&A

What is the purpose of a Clamp Meter? (Module 1)

A clamp meter is used to measure current in a live conductor without breaking the circuit. It uses the principle of magnetic induction.

Difference between MCB and RCCB? (Module 2)

MCB protects against overcurrent and short circuits. RCCB protects against earth leakage current and prevents electric shocks to humans.

Why is a resistor used with an LED? (Module 3)

LEDs have low internal resistance. A series resistor is used to limit the current flowing through the LED to prevent it from burning out.

What are the two main functions in an Arduino sketch? (Module 4)

1. `setup()`: Runs once to initialize settings. 2. `loop()`: Runs repeatedly to control the logic.

Practice Model Paper

(Non-official practice guide)

Part A: Write-up (10 Marks)

Write the aim, circuit diagram, and procedure for constructing a Bridge Rectifier on a breadboard.

Part B: Execution (10 Marks)

Wire up a single-phase ceiling fan with a regulator and demonstrate its speed control.

Part C: Arduino (10 Marks)

Write a code to blink an LED with a 2-second delay and upload it to the board.

Part D: Viva (10 Marks)

Explain the working of an RCCB.

Quick Revision & Checklist

Key Points

- Always check continuity before powering a circuit.
- Phase wire must always go through the switch.
- Use 1N4007 diodes for general rectification.
- Arduino pins can provide max 40mA (keep it safe).

Exam Checklist

- ☐ Lab Record Certified?
- ☐ Multimeter Probes checked?
- ☐ Soldering Iron cleaned?

Glossary

Breadboard

A construction base for prototyping electronics.

NCV

Non-Contact Voltage detection.

BLDC

Brushless Direct Current motor.

IDE

Integrated Development Environment (software for coding).

References

- *Fundamentals of Electrical and Electronics Engineering* - Susan S. Mathew, Saji T. Chacko.
 - *Basic Electrical Lab Manual* - S. K. Sahdev.
 - *Electronics Lab Manual* - K.A. Navas.
 - *Arduino made Simple* - Ashwin Pajankar.
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